

Azure, AWS and GCP have multiple compute models and instance types, we may not know if we are using the right compute size and model. If we can optimise the compute model considering performance and technical requirements, we can manage costs better.

Open source cloud cost management tools

Kube-Downscaler

Kube-Downscaler is an open source framework written in Python, which can be used to handle automatic shutdowns or set the uptime for Kubernetes worker pods based on usage patterns (for example, shutdown during the weekend and set uptime during office hours).

This framework runs on a single pod in the Kubernetes cluster and can work with Azure (AKS), AWS (EKS) and GCP (GKE). It has a configuration setup that can be given for the shutdown time period or uptime running period. It can be scheduled to run once or regularly for a given period (monthly, yearly).

Alternatively, we can use Kubernetes-Ops, which is another open source framework for handling end-to-end Kubernetes clusters. It can create pods/clusters, and manage, deploy, scale and monitor them in a single pane of glass.

Kubecost

Though cloud cost governance aims at managing compute and storage services to a large extent, managing container costs is also a challenging exercise.

Kubecost helps to get monthly spends, costs for name space level, and deployment resource costs; it identifies cost efficiency in a Kubernetes clustered environment. It comes in both a free-to-use open source model (where there are limitations to using it such as a single cluster, metric retention period) and a commercial model.

We can get details of underutilised or over-provisioned resources (nodes and pods, and abandoned resources) based on

usage footprints. The major features of Kubecost are cost allocation across nodes and pods, unified cost monitoring with integrated dashboard, cost optimisation insights to identify under-provisioned or over-provisioned cluster resources, and cost alerts as well as notifications for anomalous behaviour in resource usage.

Kubecost can be used with Azure Kubernetes Service (AKS), Amazon's Elastic Kubernetes Service (EKS) and Fargate, Google Kubernetes Engine (GKE) and other Kubernetes platforms in private cloud environments.

Integrated FinOps toolchain solution

FinOps is more of a culture adoption and is driven by the three-dimensional approach for cloud cost management — people (cost administration, team structure, training), processes (cost levers, maturity models, metrics definitions) and tools (cost management services, third party tools, cost advisory services).

FinOps is a collaborative activity between the finance/procurement team, technical IT operations team, and business management (COO, CIO). It can be developed as an agnostic framework for implementation in a multi-cloud environment (public/

private/hybrid). It can also be easily integrated with third party tools for flexible implementation. For example, developing reports and dashboards could be made flexible with this integration of third-party tools.

Future trends in FinOps adoption

The adoption of FinOps is maturing in most organisations that have substantial cloud consumption as they need processes and procedures to manage, monitor, allocate and forecast cloud spending. With the introduction of FinOps, organisations will have deeper insights into the resources available or reserved, which will promote appropriate procurement of cloud resources and avoid unnecessary spend in the first place.

Choosing between the right FinOps platform from cloud services providers (Azure Cost Management, AWS Cost Explorer and GCP Cost Tools), third party options or in-house development is not an easy task. Factors like variability in size and usage of cloud resources, fast-changing technology, and the advent of new cloud use cases complicate the process of automating cost optimisation. Building a governance culture and mindset is vital, but not always easy. **END** 🐧

References

- [1] https://blogs.gartner.com/lydia_leong/2021/11/24/the-cloud-budget-overrun-rainbow-of-flavors/
- [2] <https://www.kubecost.com/features>
- [3] <https://opengov.com/article/using-kubernetes-downscaler-for-time-of-day-and-day-of-week-automation>
- [4] <https://www.apptio.com/blog/where-finops-and-tbm-intersect/>
- [5] <https://www.cloudtp.com/doppler/cloud-economics-getting-bigger-picture/>
- [6] <https://www.softchoice.com/blogs/cloud-migration-adoption-management/whitepaper-finops-the-future-of-cloud-economics>

By: Dr Anand Nayyar and Dr Magesh Kasthuri

Dr Anand Nayyar is a PhD in wireless sensor networks and swarms intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning and cyber security.

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.